3.3 Practice SOLUTIONS

1. Write each of the following in terms of the co-function identity:

(a) $\sin\left(\frac{5\pi}{8}\right)$

$\frac{\pi}{2} - x = \frac{5\pi}{8}$

$\frac{\pi}{2} - \frac{5\pi}{8} = x$

$-\pi/8 = x$

$= \cos(-\pi/8)$

(b) $\csc\left(\frac{5\pi}{18}\right)$

$\frac{\pi}{2} - x = \frac{5\pi}{18}$

$\frac{\pi}{2} - \frac{5\pi}{18} = x$

$\frac{2\pi}{9} = x$

$= \sec\left(\frac{2\pi}{9}\right)$

(c) $\cos\left(\frac{\pi}{9}\right)$

$\frac{\pi}{2} - x = \frac{\pi}{9}$

$\frac{\pi}{2} - \frac{\pi}{9} = x$

$\frac{7\pi}{18} = x$

$= \sin\left(\frac{7\pi}{18}\right)$

(d) $\cos\left(\frac{7\pi}{36}\right)$

$\frac{\pi}{2} - x = \frac{7\pi}{36}$

$\frac{\pi}{2} - \frac{7\pi}{36} = x$

$\frac{11\pi}{36} = x$

$= \sin\left(\frac{11\pi}{36}\right)$

2. Find the exact value of special angles using equivalent trigonometric expressions.

a) $\cot\left(\frac{-5\pi}{3}\right)$

$= \cot\left(-2\pi + \frac{\pi}{3}\right)$

$= \cot\left(\frac{\pi}{3}\right)$

$= \frac{1}{\sqrt{3}}$

$= \sqrt{3}/3$

b) $\csc\left(\frac{4\pi}{3}\right)$

$= \csc\left(\pi + \frac{\pi}{3}\right)$

$= -\csc\left(\frac{\pi}{3}\right)$

$= -\frac{2}{\sqrt{3}}$

$= -\frac{2\sqrt{3}}{3}$

c) $\sin\left(\frac{10\pi}{3}\right)$

$= \sin\left(3\pi + \frac{\pi}{3}\right)$

$= -\sin\left(\frac{\pi}{3}\right)$

$= -\frac{\sqrt{3}}{2}$

d) $\sec\left(\frac{2\pi}{3}\right)$

$= \sec\left(\pi - \frac{\pi}{3}\right)$

$= -\sec\left(\frac{\pi}{3}\right)$

$= -2$

e) $\sin\left(\frac{-7\pi}{6}\right)$

$= \sin\left(-\pi - \frac{\pi}{6}\right)$

$= \sin\left(\frac{\pi}{6}\right)$

$= \frac{1}{2}$

f) $\cot\left(\frac{\pi}{2} - x\right)\cot(x)$

$= \tan(x) \cdot \frac{1}{\tan(x)}$

$= 1$

3. Express each of the following as a single trigonometric ratio, <u>and then</u> evaluate the ratio.

a) $\sec^2\left(\frac{7\pi}{6}\right)\cot^2\left(\frac{7\pi}{6}\right)$

$= \frac{1}{\cancel{\cos^2\left(\frac{7\pi}{6}\right)}} \cdot \frac{\cancel{\cos^2\left(\frac{7\pi}{6}\right)}}{\sin^2(7\pi/6)}$

$= \frac{1}{\sin^2(7\pi/6)}$

$= \csc^2(7\pi/6)$

$= \left[\csc\left(\pi + \frac{\pi}{6}\right)\right]^2$

$= \left[-\csc\left(\frac{\pi}{6}\right)\right]^2$

$= (-2)^2$

$= 4$

b) $\csc^2\left(\frac{3\pi}{4}\right)\tan^2\left(\frac{3\pi}{4}\right)$

$= \frac{1}{\cancel{\sin^2\left(\frac{3\pi}{4}\right)}} \cdot \frac{\cancel{\sin^2\left(\frac{3\pi}{4}\right)}}{\cos^2\left(\frac{3\pi}{4}\right)}$

$= \frac{1}{\cos^2\left(\frac{3\pi}{4}\right)}$

$= \sec^2\left(\frac{3\pi}{4}\right)$

$= \left[\sec\left(\pi - \frac{\pi}{4}\right)\right]^2$

$= \left[-\sec\left(\frac{\pi}{4}\right)\right]^2$

$= (-\sqrt{2})^2$

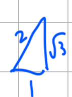
$= 2$

c) $\cos\left(\frac{4\pi}{3}\right) - \csc\left(\frac{5\pi}{6}\right)$

$= \cos\left(\pi + \frac{\pi}{3}\right) - \csc\left(\frac{\pi}{2} + \frac{\pi}{3}\right)$

$= -\cos\left(\frac{\pi}{3}\right) - \sec\left(\frac{\pi}{3}\right)$

$= -\cos\left(\frac{\pi}{3}\right) - \frac{1}{\cos\left(\frac{\pi}{3}\right)}$



$= -\frac{1}{\cos\left(\frac{\pi}{3}\right)}\left[\cos^2\frac{\pi}{3} - 1\right]$

$= -\frac{1}{\cos\left(\frac{\pi}{3}\right)}\left[\frac{1}{4} + 1\right]$

$= \frac{-5}{4\cos\left(\frac{\pi}{3}\right)}$

$= \frac{-5}{4(1/2)}$

$= -5/2$

4. Determine a value for c such that $\tan\left(2c+\frac{\pi}{3}\right)-\cot\left(3c-\frac{\pi}{4}\right)=0$.

$\tan\left(2c+\frac{\pi}{3}\right)=\cot\left(3c-\frac{\pi}{4}\right)$

By the identity $\tan\left(\frac{\pi}{2}-x\right)=\cot(x) \Rightarrow \frac{\pi}{2}-\left(2c+\frac{\pi}{3}\right)=3c-\frac{\pi}{4}$

$$\frac{\pi}{6}-2c=3c-\frac{\pi}{4}$$

$$\frac{5\pi}{12}=5c$$

$$\frac{\pi}{12}=c$$

5. Determine a value for θ that satisfies $\sin\left(2\theta-\frac{\pi}{3}\right)=\cos\left(\frac{5\pi}{4}-3\theta\right)$.

By the identity $\sin\left(\frac{\pi}{2}-x\right)=\cos x \Rightarrow \frac{\pi}{2}-\left(2\theta-\frac{\pi}{3}\right)=\frac{5\pi}{4}-3\theta$

$$\frac{5\pi}{6}-2\theta=\frac{5\pi}{4}-3\theta$$

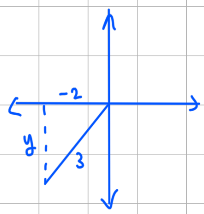
$$\theta=\frac{5\pi}{12}$$

6. Given $\sin\left(x+\frac{3\pi}{2}\right)=\frac{2}{3}$ where $\pi < x < \frac{3\pi}{2}$, determine the exact value of $\sin(x)$.

$\sin\left(x+\frac{3\pi}{2}\right)=-\cos x$

Solve: $-\cos x = 2/3$

$\cos x = -2/3$



$x^2+y^2=r^2$

$4+y^2=9$

$y=\sqrt{5}$

$\therefore \sin x = -\frac{\sqrt{5}}{3}$

7. Simplify.

(a) $\cos(-\theta)+\cos(180^\circ-\theta)-\cos(180^\circ+\theta)$

$= \cos\theta + (-\cos\theta) - (-\cos\theta)$

$= \cos\theta - \cos\theta + \cos\theta$

$= \cos\theta$

(b) $\dfrac{\cos(\pi+x)+\sin(\pi-x)}{\sin(2\pi-x)-\cos(\pi-x)}$

$= \dfrac{-\cos x + \sin x}{-\sin x - (-\cos x)}$

$= \dfrac{\sin x - \cos x}{-\sin x + \cos x}$

$= \dfrac{-(\cancel{-\sin x+\cos x})}{\cancel{-\sin x+\cos x}}$

$= -1$

(c) $\sin(\pi+x)\cos\left(\frac{\pi}{2}-x\right)-\cos(\pi+x)\sin\left(x+\frac{3\pi}{2}\right)$

$= -\sin x \cdot \sin x - (-\cos x)(-\cos x)$

$= -\sin^2 x - \cos^2 x$

$= -\sin^2 x - (1-\sin^2 x)$

$= -1$

(d) $\dfrac{\cos(-x)\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{3\pi}{2}-x\right)\csc(x)}{\sin\left(\frac{\pi}{2}+x\right)-\cos\left(x-\frac{\pi}{2}\right)+\sin\left(\frac{3\pi}{2}-x\right)}$

$= \dfrac{\cos x(\cos x) + (-\cancel{\sin x}) \cdot \frac{1}{\cancel{\sin x}}}{\cos x - (-\sin x) + (-\cos x)}$

$= \dfrac{\cos^2 x - 1}{\sin x}$

$= \dfrac{\sin^2 x}{\sin x}$

$= \sin x$

8. Fill in the blanks with the appropriate trig function name:

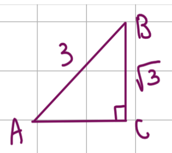
(a) $\sin\frac{2\pi}{3} = \underline{\ \cos\ }\left(\frac{-\pi}{6}\right)$ $\qquad \frac{\pi}{2}-\frac{2\pi}{3} = \frac{3\pi}{6}-\frac{4\pi}{6} = -\frac{\pi}{6}$

(b) $\underline{\ \cos\ }\ \frac{11\pi}{60} = \sin\frac{19\pi}{60}$ $\qquad \frac{\pi}{2}-\frac{11\pi}{2} = \frac{30\pi}{2}-\frac{11\pi}{2} = \frac{19\pi}{2}$

(c) $\cos\frac{7\pi}{18} = \dfrac{1}{\underline{\ \csc\ }\ \frac{\pi}{9}}$ $\qquad \frac{\pi}{2}-\frac{7\pi}{18} = \frac{9\pi}{18}-\frac{7\pi}{18} = \frac{2\pi}{18} = \frac{\pi}{9}$

9. For right triangle ABC:

(a) If $\sin A = \frac{\sqrt{3}}{3}$, what is the value of cos B?



$\cos B = \frac{\sqrt{3}}{3}$

(b) If cos A=0.109, what is $\sin\left(\frac{\pi}{2}-A\right)$?

$\sin\left(\frac{\pi}{2}-A\right) = \cos A = 0.109$

(c) If $\cos\frac{11\pi}{180} = 0.9816$, what is $\sin\frac{79\pi}{180}$?

$\frac{\pi}{2} - \frac{11\pi}{180} = \frac{90\pi}{180} - \frac{11\pi}{180} = \frac{79\pi}{180}$

$\sin\left(\frac{79\pi}{180}\right) = \sin\left(\frac{\pi}{2}-\frac{11\pi}{180}\right) = \cos\left(\frac{11\pi}{180}\right) = 0.9816$